

4.2

Spindle Warm-Up

If your machine's spindle has been idle for more than (4) days, run the spindle warm-up program before you use the machine. This program brings the spindle up to speed slowly to distribute the lubrication and let the spindle reach a stable temperature.

Your machine includes a 20-minute warm-up program (009220) in the program list. If you use the spindle at consistent high speeds, you should run this program every day.

4.3

Device Manager ([LIST PROGRAM])

You use the device manager ([LIST PROGRAM]) to access, save, and manage data on the CNC control, and on other devices attached to the control. You also use the device manager to load and transfer programs between devices, set your active program, and back up your machine data.

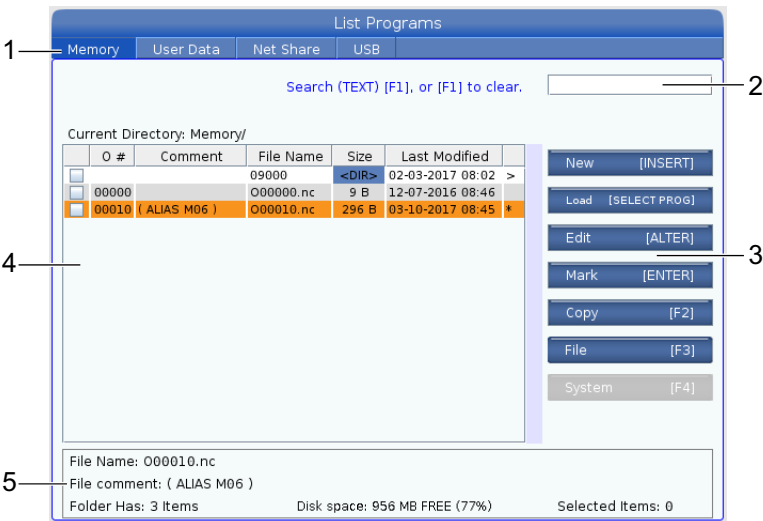
In the tabbed menu at the top of the display, the device manager ([LIST PROGRAM]) shows you only the available memory devices. For example, if you do not have a USB memory device connected to the control pendant, the tabbed menu does not show a **USB** tab. For more information on navigating tabbed menus, refer to page 65.

The device manager ([LIST PROGRAM]) shows you the available data in a directory structure. At the root of the CNC control are the available memory devices in a tabbed menu. Each device can contain combinations of directories and files, many levels deep. This is similar to the file structure you find in common personal computer operating systems.

4.3.1 Device Manager Operation

Press **[LIST PROGRAM]** to access the device manager. The initial device manager display shows the available memory devices in a tabbed menu. These devices can include machine memory, the User Data directory, USB memory devices connected to the control, and files available on the connected network. Select a device tab to work with the files on that device.

F4.2: Device Manager Initial Screen Example: [1] Available Device Tabs, [2] Search Box, [3] Function Keys, [4] File Display, [5] File Comments (only available in **Memory**).



Use the cursor arrow keys to navigate the directory structure:

- Use the **[UP]** and **[DOWN]** cursor arrow keys to highlight and interact with a file or a directory in the current root or directory.
- Roots and directories have a right-hand arrow character (>) in the far-right column of the file display. Use the **[RIGHT]** cursor arrow key to open a highlighted root or directory. The display then shows the contents of that root or directory.
- Use the **[LEFT]** cursor arrow key to return to the previous root or directory. The display then shows the contents of that root or directory
- The CURRENT DIRECTORY message above the file display tells you where you are in the directory structure; for example: *MEMORY/CUSTOMER 11/NEW PROGRAMS* shows that you are in the subdirectory **NEW_PROGRAMS** inside the directory **CUSTOMER 11**, in the root of **MEMORY**.

4.3.2 File Display Columns

When you open a root or directory with the **[RIGHT]** cursor arrow key, the file display shows you a list of the files and directories in that directory. Each column in the file display has information about the files or directories in the list.

F4.3: Program/Directory List Example

Current Directory: Memory/						
	O #	Comment	File Name	Size	Last Modified	
☐			TEST	<DIR>	2015/11/23 08:54	>
☐			programs	<DIR>	2015/11/23 08:54	>
☐	00010		O00010.nc	130 B	2015/11/23 08:54	
☒	00030		O00030.nc	67 B	2015/11/23 08:54	*
☐	00035		O00035.nc	98 B	2015/11/23 08:54	
☐	00045		NEXTGENTe...	15 B	2015/11/23 08:54	
☐	09001 (ALIAS M89)		O9001.nc	94 B	2015/11/23 08:54	

The columns are:

- File selection check box (no label): Press **ENTER** to toggle a check mark on and off in the box. A check mark in a box indicates that the file or directory selected for operations on multiple files (usually copy or delete).
- Program O Number (o #): This column lists the program numbers of the programs in the directory. The letter 'O' is omitted in the column data. Only available in the **Memory** tab.
- File comment (Comment): This column lists the optional program comment that appears in the first line of the program. Only available in the **Memory** tab.
- File Name (File Name): This is an optional name that the control uses when you copy the file to a memory device other than the control. For example, if you copy program o00045 to a USB memory device, the filename in the USB directory is **NEXTGENTest.nc**.
- File Size (Size): This column shows the amount of storage space that the file takes up. Directories in the list have the designation **<DIR>** in this column.



NOTE:

*This column is hidden by default, press the **[F3]** button and select **Show File Details** to display this column.*

- Date Last Modified (Last Modified): This column shows the last date and time when the file was changed. The format is YYYY/MM/DD HR:MIN.

**NOTE:**

*This column is hidden by default, press the **[F3]** button and select **Show File Details** to display this column.*

- Other Information (no label): This column gives you some information about a file's status. The active program has an asterisk (*) in this column. A letter **E** in this column means that the program is in the program editor. A greater-than symbol (>) indicates a directory. A letter **S** indicates that a directory is part of Setting 252 (refer to page **395** for more information). Use the **[RIGHT]** or **[LEFT]** cursor arrow keys to enter or exit the directory.

4.3.3 Create a New Program

Press **[INSERT]** to create a new file in the current directory. The **CREATE NEW PROGRAM** popup menu shows on the screen:

F4.4: Create New Program Popup Menu Example: [1] Program O number field, [2] File Name field, [3] File comment field.

The screenshot shows a 'Create New Program' dialog box. It has a blue header bar with the title 'Create New Program'. Inside the dialog, there are three text input fields. The first field is labeled 'O Number*' and has a '1' pointing to it. The second field is labeled 'File Name*' and has a '2' pointing to it. The third field is labeled 'File comment' and has a '3' pointing to it. Below these fields is a text prompt 'Enter an O number or file name'. At the bottom of the dialog are two buttons: 'Enter [ENTER]' and 'Exit [UNDO]'.

Enter the new program information in the fields. The **Program O number** field is required; the **File Name** and **File comment** are optional. Use the **[UP]** and **[DOWN]** cursors to move between the menu fields.

Press **[UNDO]** at any time to cancel program creation.

- **Program O number** (required for files created in Memory): Enter a program number up to (5) digits long. The control adds the letter **O** automatically. If you enter a number shorter than (5) digits, the control adds leading zeros to the program number to make it (5) digits long; for example, if you enter **1**, the control adds zeros to make it **00001**.

**NOTE:**

Do not use 009XXX numbers when you create new programs. Macro programs often use numbers in this block, and overwriting them may cause machine functions to malfunction or stop working.

- **File Name** (optional): Type a filename for the new program. This is the name the control uses when you copy the program to a storage device other than memory.
- **File comment** (optional): Type a descriptive program title. This title goes into the program as a comment in the first line with the O number.

Press **[ENTER]** to save your new program. If you specified an O number that exists in the current directory, the control gives the message *File with O Number nnnnn already exists. Do you want to replace it?* Press **[ENTER]** to save the program and overwrite the existing program, press **[CANCEL]** to return to the program name popup, or press **[UNDO]** to cancel.

4.3.4 Select the Active Program

Highlight a program in the memory directory, then press **[SELECT PROGRAM]** to make the highlighted program active.

The active program has an asterisk (*) in the far-right column in the file display. It is the program that runs when you press **[CYCLE START]** in **OPERATION:MEM** mode. The program is also protected from deletion while it is active.

4.3.5 Check Mark Selection

The check box column on the far left of the file display lets you select multiple files.

Press **[ENTER]** to place a check mark in a file's check box. Highlight another file and press **[ENTER]** again to put a check mark in that file's check box. Repeat this process until you have selected all of the files you want to select.

You can then do an operation (usually copy or delete) on all of those files at the same time. Each file that is part of your selection has a check mark in the check box. When you choose an operation, the control does that operation on all of the files with check marks.

For example, if you want to copy a set of files from the machine memory to a USB memory device, you would put a check mark on all of the files that you want to copy, then press **[F2]** to start the copy operation.

To delete a set of files, put a check mark on all of files you want to delete, then press **[DELETE]** to start the delete operation.

**NOTE:**

A check-mark selection only marks the file for further operation; it does not make the program active.



NOTE:

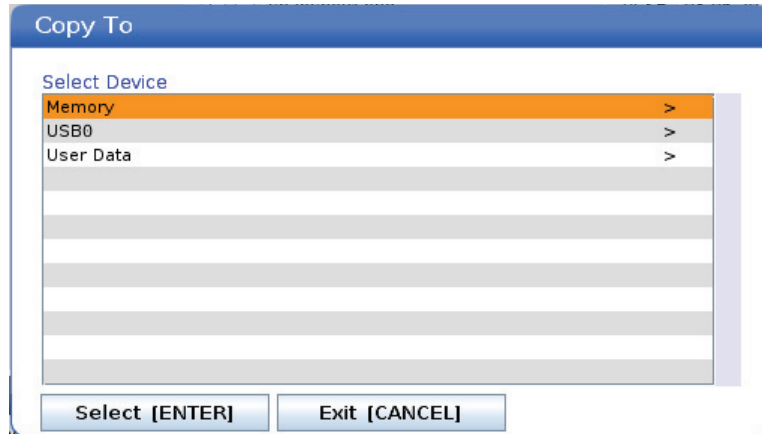
If you have not selected multiple files with check marks, the control does operations only on the currently highlighted directory or file. If you have selected files, the control does operations only on the selected files and not on the highlighted file, unless it is also selected.

4.3.6 Copy Programs

This function lets you copy programs to a device or a different directory.

1. To copy a single program, highlight it in the device manager program list and press **[ENTER]** to assign a check mark. To copy multiple programs, check-mark all the programs you want to copy.
2. Press **[F2]** to start the copy operation.
The Select Device popup appears.

F4.5: Select Device

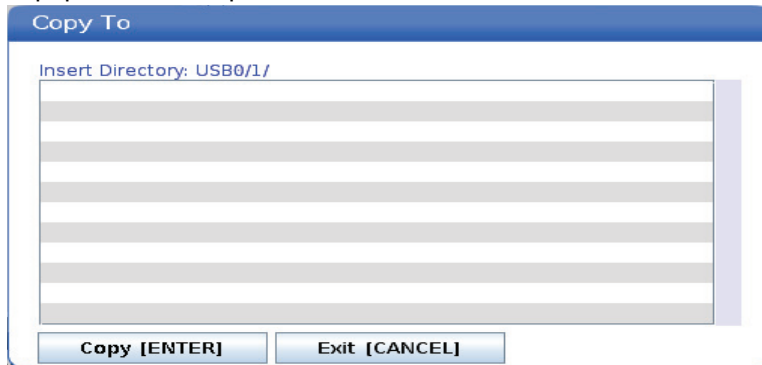


3. Use the cursor arrow keys to select the destination directory. **[RIGHT]** cursor to enter the chosen directory.

The **Insert Directory**: Copy popup menu appears.

F4.6:

Copy Popup Menu Example



4. Press **[ENTER]** to complete the copy operation, or press **[CANCEL]** to return to the device manager.

4.3.7 Edit a Program

Highlight a program, and then press **[ALTER]** to move the program into the program editor.

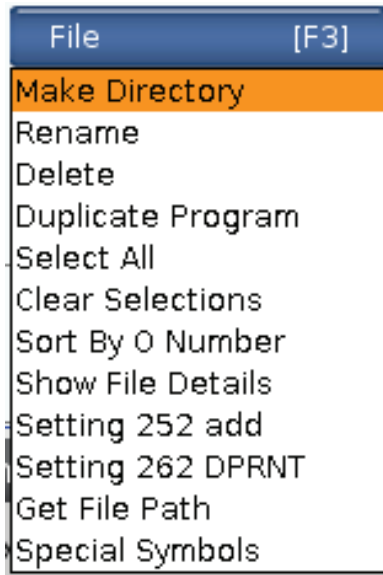
The program has the designation **E** in the far-right column of the file display list when it is in the editor, unless it is also the active program.

You can use this function to edit a program while the active program runs. You can edit the active program, but your changes do not take effect until you save the program and then select it again in the device manager menu.

4.3.8 File Commands

Press **[F3]** to access the file commands menu in the device manager. The list of options appears under the **File [F3]** drop-down menu in the device manager. Use the cursor arrow keys or the jog handle to highlight a command, and then press **[ENTER]**.

F4.7: The File Commands Menu



- **Make Directory:** creates a new subdirectory in the current directory. Type a name for the new directory, and then press **[ENTER]**.
- **Rename:** changes the name of a program. The **Rename** popup menu has the same options as the new program menu (File Name, O Number, and File Title).
- **Delete:** deletes files and directories. When you confirm the operation, the control deletes the highlighted file, or all of the files with check-mark selection.
- **Duplicate Program:** makes a copy of a file in the current location. The **Save As** popup menu asks you to specify a new program name before you can complete this operation.
- **Select All:** adds check-marks to all of the files/directories in the **Current Directory**.
- **Clear Selections:** removes check-marks from all of the files/directories in the **Current Directory**.
- **Sort By O Number:** sorts the program list by O number. Use this menu item again to sort by filename. By default, the program list is sorted by filename. Only available in the **Memory** tab.
- **Setting 252 add / Setting 252 remove:** adds or removes a custom sub-program search location to the list of locations. See Setting Up Search Locations section for more information.
- **Setting 262 DPRNT:** adds a custom destination file path for DPRNT.